Back to Stochastic Processes

The random variables in a stochastic process $\{X_n\}_{n=0}^{\infty}$ take values from a <u>common</u> <u>state</u> <u>space</u> S.

(Common to all X_n !)

The Markov Property by example:

$$P(X_4 = 5 \mid X_3 = 2, X_2 = -1, X_1 = 3, X_0 = 1)$$

$$= P(X_4 = 5 \mid X_3 = 2)$$

"Only the most recent piece of given information matters"

<u>Definition</u>: Let S be a discrete state space.

The discrete-time stochastic process $\{X_n\}_{n=0}^{\infty}$ on S is a Markov chain (process) if

$$P(X_{n+1}=j \mid X_n=i, X_{n-1}=i_{n-1}, \ldots, X_0=i_0)$$
$$= P(X_{n+1}=j \mid X_n=i)$$

for all $n=0,1,2,\ldots$ and for all $i_0,\ldots,i_{n-1},i,j \in S$ whenever these conditional probabilities are defined.

<u>Definition</u>: Let S be a discrete state space.

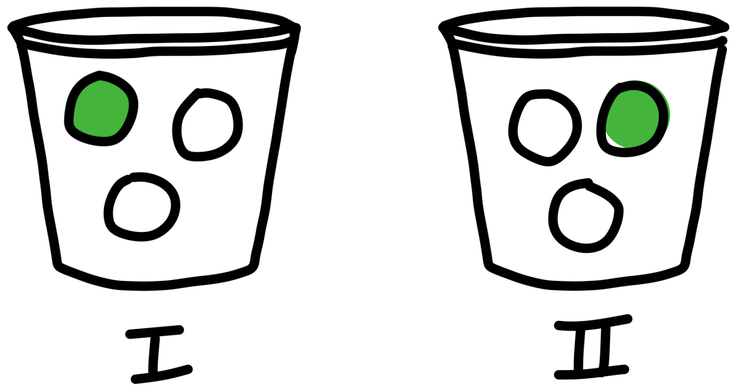
The discrete-time stochastic process $\{X_n\}_{n=0}^{\infty}$ on S is a Markov chain (process) if

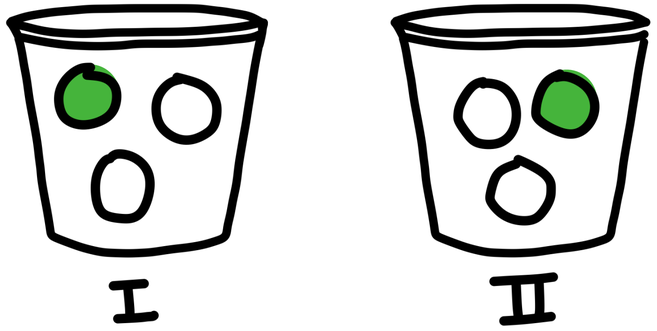
$$P(X_{n+1}=j \mid X_n=i, X_{n-1}=i_{n-1}, \ldots, X_0=i_0) = P(X_{n+1}=j \mid X_n=i)$$

"the Markov Property"

Example:

Two green balls and four white balls are put into two buckets in such a way that each bucket contains 3 balls.



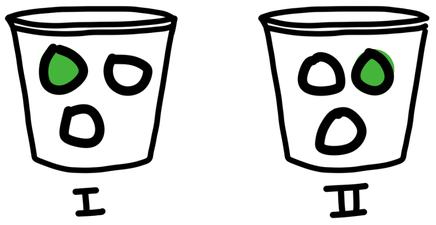


At each time step, we randomly select one ball from each bucket and interchange them.

Let X_n = # green balls in bucket I after the nth exchange

Claim: $\{X_n\}$ is a Markov chain

$S = \{0, 1, 2\}$

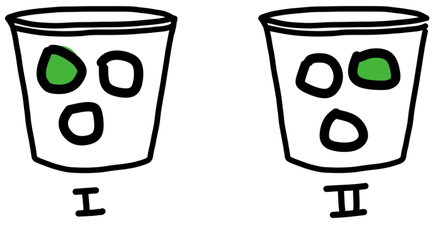


$$P(\underbrace{X_4 = 1}_{\text{next time step}} \mid \underbrace{X_3 = 1}_{\text{current time}}, \underbrace{X_2 = 2, X_1 = 1, X_0 = 0}_{\text{past history}})$$

$$= P(X_4 = 1 \mid X_3 = 1) \checkmark$$

<u>Claim:</u> $\{X_n\}$ is a Markov chain

$S = \{0, 1, 2\}$



$P(X_4 = 1 \mid X_3 = 1)$

$= P(\text{draw a G from I and G from II})$

$+ P(\text{draw a W from I and W from II})$

$\underset{\text{indep}}{=} \frac{1}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{2}{3} = \boxed{\frac{5}{9}}$

For this example, note that

$$P(X_4 = 1 \mid X_3 = 1, X_2 = 2, X_1 = 0, X_0 = 0)$$
$$\neq P(X_4 = 1 \mid X_3 = 1)$$

$\{X_n\}$ is still a Markov chain though!

"Whenever these conditional probabilities are defined"